

System Design Problems — Advanced



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Four fully worked FAANG-style system-design solutions. Each covers requirements, capacity math, API, data model, architecture, deep dives, scaling, and trade-offs.

1 1. Design a Video Streaming Service (YouTube / Netflix)

1.1 1. Requirements

Clarifying questions to ask:

- › Are we designing upload + playback, or playback-only?
- › Live streaming, or VOD (video on demand), or both?
- › What's the target geography — global CDN or single-region?
- › Should we design the recommendation engine, or just the data hooks?
- › Premium / DRM requirements?

Functional Requirements:

- › Users can upload videos (any resolution, up to ~10 GB raw).
- › Platform transcodes video to multiple resolutions (240p, 360p, 720p, 1080p, 4K).
- › Users can stream videos with adaptive bitrate (quality switches based on bandwidth).
- › Search, browse, and play videos; like, comment, subscribe.
- › Recommendation feed on homepage.

Non-Functional Requirements:

- › Availability: 99.99% (< 53 min/yr downtime).
- › Low latency playback start: < 2 s p99.
- › Upload should not block playback — async processing pipeline.
- › Globally distributed; content served from CDN edge nodes.
- › Durability: 11 nines (video assets are irreplaceable for creators).
- › Scale: YouTube-tier — 500 hours of video uploaded per minute; 1B+ DAU.

1.2 2. Capacity Estimation

Users & traffic:

```
1 DAU = 1 billion
2 Avg watch time = 30 min/day
3 Videos uploaded = 500 hours/min = 30,000 hours/min raw footage
```

Read QPS (playback requests):

```
1B users × 5 video views/day = 5B views/day
5B / 86,400 s ≈ 58,000 views/sec → ~60K QPS peak
```

Write QPS (uploads):

```
500 hours/min ÷ 60 = ~8.3 hours/sec raw upload
~100 concurrent upload streams sustained at any time (most uploads are large, slow)
```

Storage (per year):

Raw: $500 \text{ hr/min} \times 60 \text{ min/hr} \times 24 \text{ hr/day} \times 365 \text{ days} = 262,800,000 \text{ hr/yr}$
 Avg 1 hr video raw $\approx 2 \text{ GB} \rightarrow 525 \text{ PB/yr raw}$
 Transcoded: $\sim 5 \text{ resolutions} \times 0.4 \times \text{compression ratio each} = \sim 2 \times \text{raw}$
 Total $\approx 1 \text{ EB/yr}$ (YouTube stores $\sim 1 \text{ EB}$ today)

Bandwidth (outbound CDN):

60K concurrent streams $\times$ avg 3 Mbps (720p) = 180 Gbps
 Peak (2 $\times$ avg) = 360 Gbps served by CDN PoPs

Cost hint: At \$0.01/GB CDN egress, $180 \text{ Gbps} \times 3600 \text{ s} \times 24 \text{ h} = \sim \1.5 M/day — hence CDN caching is existential.

1.3 3. API Design

```

1 # Upload
2 POST /v1/videos/initiate-upload
3   Body: { title, description, tags, category }
4   Returns: { upload_id, upload_url (signed S3) }
5
6 PUT /v1/videos/upload/{upload_id}/chunk
7   Headers: Content-Range, Content-Length
8   Body: <binary chunk>
9
10 POST /v1/videos/upload/{upload_id}/complete
11   Returns: { video_id, status: "processing" }
12
13 # Playback
14 GET /v1/videos/{video_id}
15   Returns: { metadata, manifest_url (HLS/DASH), thumbnail_url }
16
17 GET /v1/videos/{video_id}/manifest.m3u8 (served by CDN)
18
19 # Discovery
20 GET /v1/feed?user_id=&page_token=
21 GET /v1/search?q=&sort=relevance&page=
22 GET /v1/videos/{video_id}/recommendations
23
24 # Engagement
25 POST /v1/videos/{video_id}/views (fire-and-forget)
26 POST /v1/videos/{video_id}/likes
27 POST /v1/videos/{video_id}/comments
  
```

1.4 4. Data Model

Metadata DB — PostgreSQL (sharded) or Spanner (global):

```

1 videos (
2   video_id      UUID PRIMARY KEY,
3   creator_id    BIGINT,
4   title         TEXT,
5   description   TEXT,
6   status        ENUM('uploading', 'processing', 'ready', 'failed'),
7   duration_s    INT,
8   raw_s3_key    TEXT,
9   created_at    TIMESTAMPTZ,
  
```

```

10  view_count      BIGINT,      -- denormalized, updated async
11  like_count      BIGINT
12 )
13
14 video_renditions (
15  video_id        UUID,
16  resolution       TEXT,      -- '1080p', '720p', ...
17  codec           TEXT,      -- 'h264', 'h265', 'av1'
18  manifest_key     TEXT,      -- S3 key for HLS/DASH manifest
19  segment_prefix   TEXT,      -- S3 prefix for .ts segments
20  size_bytes       BIGINT
21 )
22
23 channels (channel_id, owner_user_id, subscriber_count, ...)
24 subscriptions (subscriber_id, channel_id, created_at)
25 comments (comment_id, video_id, user_id, body, created_at)

```

Why PostgreSQL/Spanner? Transactional metadata (status updates, renditions) needs ACID. Spanner if truly global consistency is required; sharded Postgres otherwise.

Counters — Redis or dedicated counter service:

- › VIEW: {video_id} → HyperLogLog for approximate unique views; batched flush to Postgres.

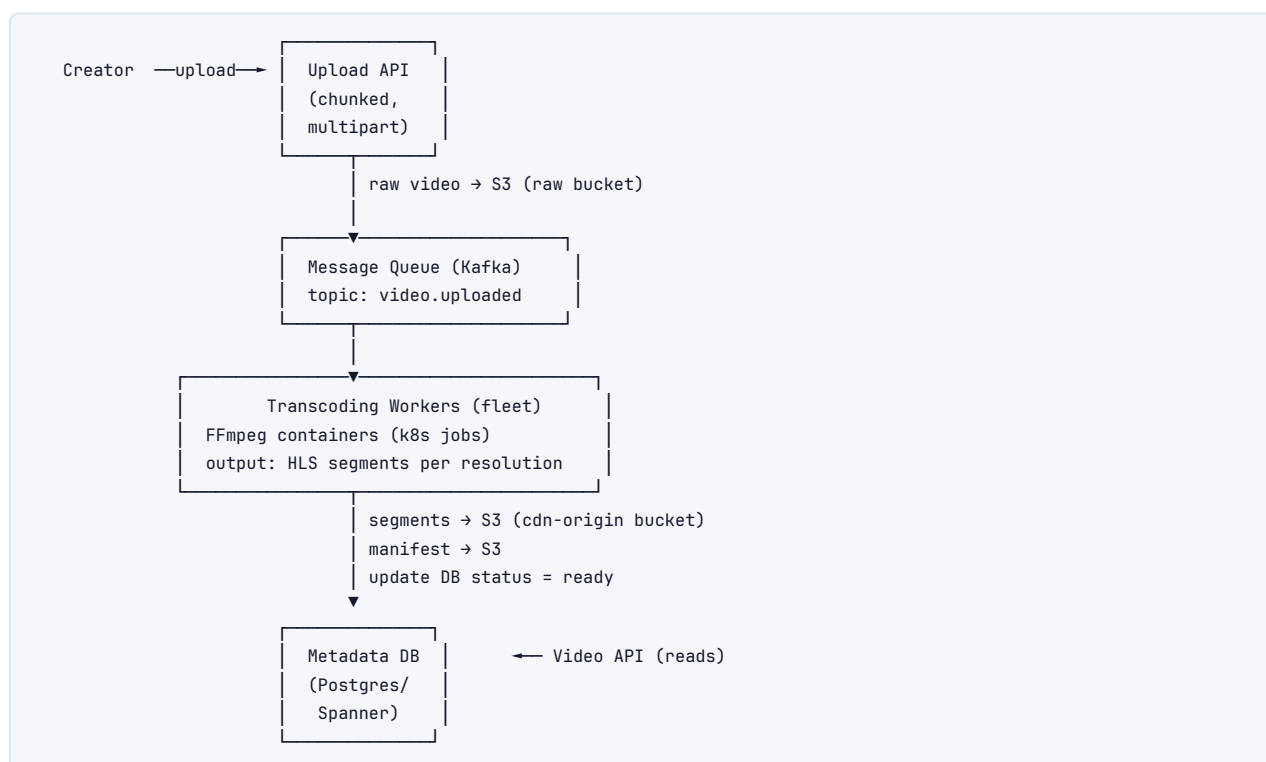
Search — Elasticsearch:

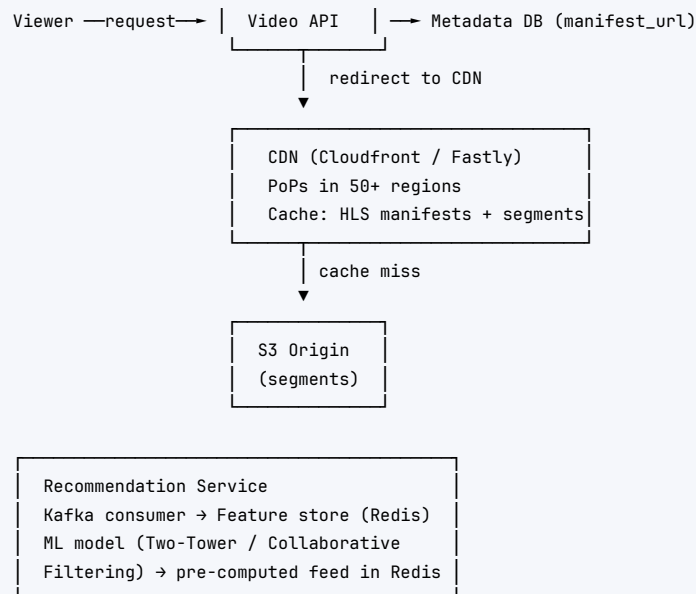
- › Index: { video_id, title, tags, transcript_excerpt, view_count, created_at }.
- › Full-text + vector embedding for semantic search.

Blob Storage — S3-compatible object store (AWS S3 / GCS):

- › Raw uploads in one bucket; transcoded segments in another bucket per region.
- › Lifecycle policy: raw deleted after 30 days if status=ready.

1.5 5. High-Level Architecture





1.6 6. Deep Dives

6.1 Upload Pipeline & Transcoding

Chunked Upload:

1. Client calls `/initiate-upload` → receives a pre-signed S3 multipart upload URL.
2. Client uploads 5–10 MB chunks in parallel (S3 multipart supports up to 10K parts).
3. Client calls `/complete` → API assembles the multipart object in S3.
4. API publishes `{ video_id, s3_key }` to Kafka topic `video.uploaded`.

Transcoding Workers:

- › Kubernetes Jobs pulled from Kafka; each worker pulls raw video, runs FFmpeg.
- › **Parallel transcoding:** one worker per resolution, or one worker per video with all resolutions in parallel child processes.
- › Output: HLS (`.m3u8` master manifest + `.ts` segment files per resolution, default 6 s segments).
- › Also output a DASH MPD manifest for Chrome/Android compatibility.
- › Store thumbnails (extracted at T=5s, 30s, 60s) in S3.
- › Update `video_renditions` table and set `status = 'ready'`.

Resolution ladder (YouTube standard):

2160p (4K)	→ bitrate ~20 Mbps
1080p	→ ~8 Mbps
720p	→ ~5 Mbps
480p	→ ~2.5 Mbps
360p	→ ~1 Mbps
240p	→ ~0.5 Mbps

6.2 CDN Distribution

- › S3 origin sits behind CDN (e.g., CloudFront with 400+ PoPs).
- › HLS segments are tiny (< 1 MB each, 6 s) — extremely cache-friendly.
- › **Cache TTL:** segments are immutable → TTL = ∞ (or 30 days). Manifests refreshed every 60 s.
- › **Geo-routing:** DNS returns nearest PoP IP via Anycast/Route 53 Latency Routing.

- **Cache hit ratio target:** > 95% for popular videos. Long-tail videos miss to origin (cheap S3 GET).

6.3 Adaptive Bitrate Streaming --- HLS vs DASH

Feature	HLS (Apple)	DASH (MPEG)
Standard	Apple proprietary → RFC 8216	MPEG open standard
Container	MPEG-TS or fMP4	fMP4 (CMAF)
iOS/Safari	Native support	Requires MSE (not on older iOS)
Android/Chrome	Via MSE	Native support
DRM	FairPlay	Widevine / PlayReady
Latency (live)	6–30 s (LLHLS: 1–3 s)	2–8 s (CMAF chunked)
Adoption	YouTube, Netflix (backup)	Netflix (primary), Disney+

Key decision: Serve both HLS and DASH manifests from the same fMP4 segments (CMAF). This halves storage vs maintaining two separate segment sets.

6.4 Storage Tiers

```
Hot (< 7 days old) → S3 Standard (fast GET, higher cost)
Warm (7–90 days) → S3 Infrequent Access (30% cheaper)
Cold (> 90 days, < 1M views) → S3 Glacier Instant Retrieval
Archive (> 1yr, < 10K views) → S3 Glacier Deep Archive (pennies/GB)
```

6.5 Recommendations (brief)

- **Candidate generation:** Two-Tower model; user embedding × video embedding → top-K ANN (FAISS / ScaNN).
- **Ranking:** LightGBM or DNN re-ranker on features: CTR, watch time, freshness, diversity.
- **Serving:** Pre-compute homepage feed into Redis (TTL 30 min); real-time session signals update weights.
- **Cold start:** New videos bootstrapped with category/tag-based rules.

1.7 7. Bottlenecks & Scaling

Bottleneck	Mitigation
Transcoding backlog	Auto-scale k8s transcoding pods on Kafka consumer lag metric
Metadata DB hot rows (view counts)	Counter service with Redis; async batch-write to DB
CDN cache misses for new videos	Pre-warm CDN by pushing manifest URL to CDN after transcoding
S3 request throttling	Randomize S3 key prefixes (avoid date-prefix hotspot)
Recommendation latency	Pre-compute + cache feeds; async refresh
Single-region failure	Multi-region S3 replication; CDN absorbs reads; DB cross-region replica

1.8 8. Trade-offs Summary

Decision	Option A	Option B	Choice
Transcoding: per-video vs sharded	Single big VM per video	K8s pod per resolution	K8s pods (parallelism, cost)
Metadata DB	MySQL sharded	Spanner / CockroachDB	Spanner if global consistency needed
Streaming format	HLS-only	HLS + DASH (CMAF)	Both (broadest device coverage)
Recommendation serving	Real-time inference	Pre-computed feed	Pre-computed (latency wins)
Storage tiering	All S3 Standard	Tiered (hot/warm/cold)	Tiered (10× cost reduction)

Interview takeaways:

- › **The upload path and the playback path are entirely decoupled** — async Kafka pipeline.
- › **CDN cache hit ratio is the single most important scaling lever** for a video platform.
- › **CMAF lets you serve HLS and DASH from the same segments** — always mention this.
- › Transcoding is CPU-bound; scale it independently with Kafka consumer-lag autoscaling.

2. Design a Ride-Hailing Service (Uber / Lyft)

2.1 1. Requirements

Clarifying questions:

- › Real-time ride-matching, or scheduled rides too?
- › Single city or global deployment?
- › Do we need the driver payout/financial system?
- › Surge pricing — dynamic or simple multipliers?
- › Bike/scooter/autonomous vehicles, or just cars?

Functional Requirements:

- › Riders request a ride from location A to B.
- › System matches rider to nearby available driver in real time.
- › Driver and rider track each other on a live map during the trip.
- › Compute fare estimate upfront; charge on trip completion.
- › Surge pricing during high-demand periods.
- › Driver/rider rating after trip.

Non-Functional Requirements:

- › Matching latency < 3 s p99 from rider request to driver offer sent.
- › Location update ingestion: 1–5 s refresh from every active driver.
- › Availability: 99.99% (5 min/month downtime is unacceptable during peak hours).
- › Consistency: eventual for location data; strong for trip state (no double-booking a driver).
- › Global scale: 5M+ concurrent drivers; 10M+ concurrent riders during peak.

2.2 2. Capacity Estimation

```

1 Active drivers (peak global) = 5 million
2 Location update interval = 4 s
3 Location update QPS = 5,000,000 / 4 = 1,250,000 writes/sec
4
5 Active riders requesting rides = 500,000 concurrent
6 Ride requests QPS = 500,000 / 10s avg session = 50,000 QPS
7
8 Trips per day = 25 million (Uber ~20M/day)
9 Trip data per trip = ~2 KB metadata + GPS trace ~50 KB
10 Storage/yr = 25M × 52 KB × 365 = ~475 TB/yr
11
12 Location store size (in-memory):
13 5M drivers × (lat, lon, heading, timestamp, driver_id) = ~200 bytes each
14 = 5M × 200 B = 1 GB — fits in a large Redis cluster

```

2.3 3. API Design

```

1 # Rider
2 POST /v1/rides/estimate
3   Body: { pickup_lat, pickup_lon, dest_lat, dest_lon, ride_type }
4   Returns: { eta_s, fare_estimate, surge_multiplier }
5
6 POST /v1/rides/request
7   Body: { pickup_lat, pickup_lon, dest_lat, dest_lon, ride_type }
8   Returns: { ride_id, status: "searching" }
9
10 GET /v1/rides/{ride_id}
11   Returns: { status, driver_id, driver_eta_s, driver_location }
12
13 DELETE /v1/rides/{ride_id} (cancel)
14
15 # Driver
16 PUT /v1/drivers/{driver_id}/location
17   Body: { lat, lon, heading, speed, timestamp }
18
19 PUT /v1/drivers/{driver_id}/status
20   Body: { status: "available" | "busy" | "offline" }
21
22 GET /v1/drivers/{driver_id}/ride-offers (long-poll or WebSocket push)
23 POST /v1/ride-offers/{offer_id}/accept
24 POST /v1/ride-offers/{offer_id}/reject
25
26 # Shared real-time (WebSocket)
27 WS /v1/ws/ride/{ride_id} (bidirectional: location pings, status events)

```

2.4 4. Data Model

Drivers (PostgreSQL):

```

1 drivers (
2   driver_id      BIGINT PRIMARY KEY,
3   name          TEXT,
4   vehicle_id    BIGINT,

```

```

5   rating          NUMERIC(3,2),
6   current_status  ENUM('available','busy','offline'),
7   city_id         INT
8 )

```

Rides (PostgreSQL sharded by ride_id):

```

1 rides (
2   ride_id          UUID PRIMARY KEY,
3   rider_id         BIGINT,
4   driver_id        BIGINT,
5   status           ENUM('requested','searching','driver_assigned','in_progress','completed','cancelled'),
6   pickup_lat       DOUBLE,
7   pickup_lon       DOUBLE,
8   dest_lat         DOUBLE,
9   dest_lon         DOUBLE,
10  requested_at      TIMESTAMPTZ,
11  matched_at       TIMESTAMPTZ,
12  completed_at     TIMESTAMPTZ,
13  fare_estimate     NUMERIC(10,2),
14  final_fare       NUMERIC(10,2),
15  surge_mult       NUMERIC(4,2)
16 )

```

Driver Locations (Redis / in-memory geo store):

```

1 Key:   DRIVER_LOC:{driver_id}
2 Value: { lat, lon, heading, updated_at }
3 TTL:   30 s (stale if driver stops sending heartbeats)
4
5 Geo index: Redis GEOADD city:{city_id}:drivers {lon} {lat} {driver_id}
6           Redis GEORADIUS for nearest-N lookup

```

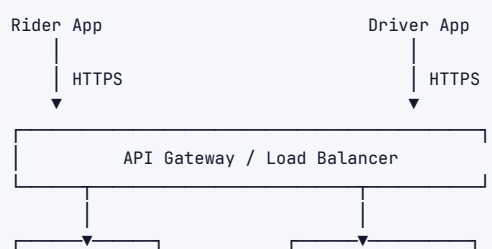
Trip GPS traces (Cassandra or TimescaleDB):

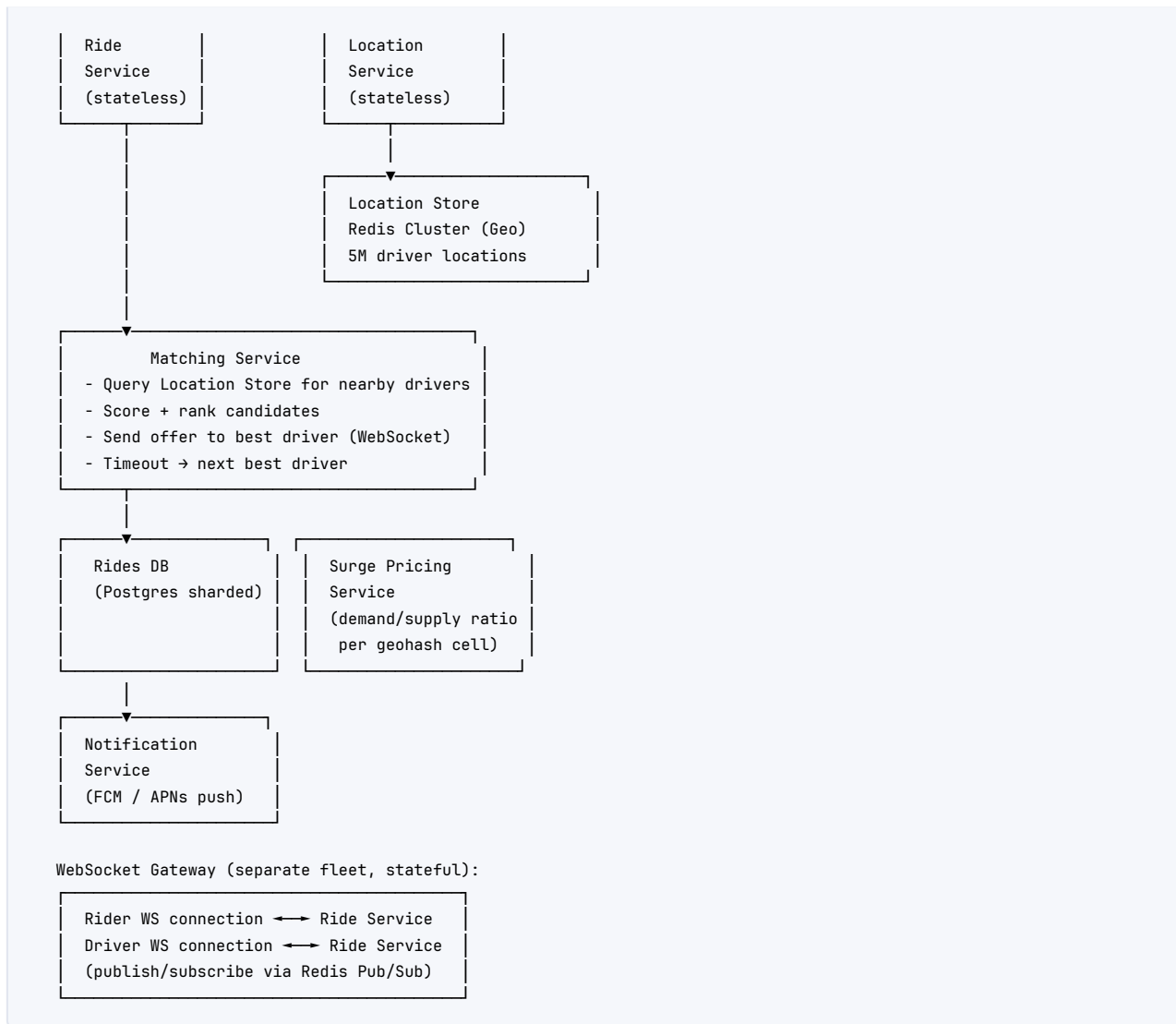
```

trip_locations (
  ride_id  UUID,
  ts       TIMESTAMPTZ,
  lat      DOUBLE,
  lon      DOUBLE,
  PRIMARY KEY (ride_id, ts)
) - append-only, time-series access pattern

```

2.5 5. High-Level Architecture





2.6 6. Deep Dives

6.1 Geospatial Indexing --- Geohash vs Quadtree vs S2

Feature	Geohash	Quadtree	Google S2
Structure	Base-32 string, hierarchical	Tree of rectangular cells	Hilbert-curve cells on sphere
Precision control	String length (1–12 chars)	Tree depth	Level (0–30)
Query type	String prefix match	Tree traversal	Cell union + covering
Edge distortion	Yes (poles, antimeridian)	Yes (aspect ratio varies)	Minimal (spherical geometry)
Neighbor lookup	8 neighbors (manual computation)	Tree sibling traversal	S2::GetEdgeNeighbors
Used by	MongoDB, Elasticsearch, Redis	Older game engines	Uber (H3), Google Maps, Foursquare
Complexity	Simple	Moderate	Higher (library dependency)

Key decision: Use Geohash (precision level 6 $\approx$ 1.2 km cells) for driver lookup.

- › Each driver is stored in Redis with GEOADD (which internally uses Geohash).

- › On ride request: GEORADIUS pickup_lat pickup_lon 3km → list of driver IDs → filter by status=available.
- › **Why not quadtree?** Redis GEODATA gives us geohash for free; quadtree requires custom in-memory structure.
- › **Why not S2?** Overkill for matching; S2 shines in complex polygon coverage (e.g., Uber H3 for analytics).

6.2 Driver-Rider Matching

Algorithm:

1. GEORADIUS call returns N available drivers within 3 km (expand to 5 km if < 3 results).
2. Score each driver:

```
1 score = w1 × (1 / ETA_to_pickup) + w2 × driver_rating + w3 × acceptance_rate
```

3. Send offer to top-scored driver. Wait 10 s.
4. If rejected/timeout → send to next driver. Max 5 tries before returning "no drivers available."
5. On acceptance → atomically set `rides.driver_id, rides.status = driver_assigned, drivers.current_status = busy` (Postgres transaction).

Preventing double-assignment: Use Postgres `SELECT FOR UPDATE` or Redis distributed lock (`SET driver:{id}:lock NX EX 15`) before assigning.

6.3 Real-Time Location Updates

- › Drivers send GPS ping every 4 s via HTTP/2 (or WebSocket for persistent connection).
- › **Location Service** writes to Redis GEOADD ($O(\log N)$) — 1.25M writes/sec requires Redis Cluster with 10–20 shards.
- › **Sharding:** partition by `city_id` or geohash prefix (level 2 $\approx$ 630 km cells → ~100 world cells → manageable shards).

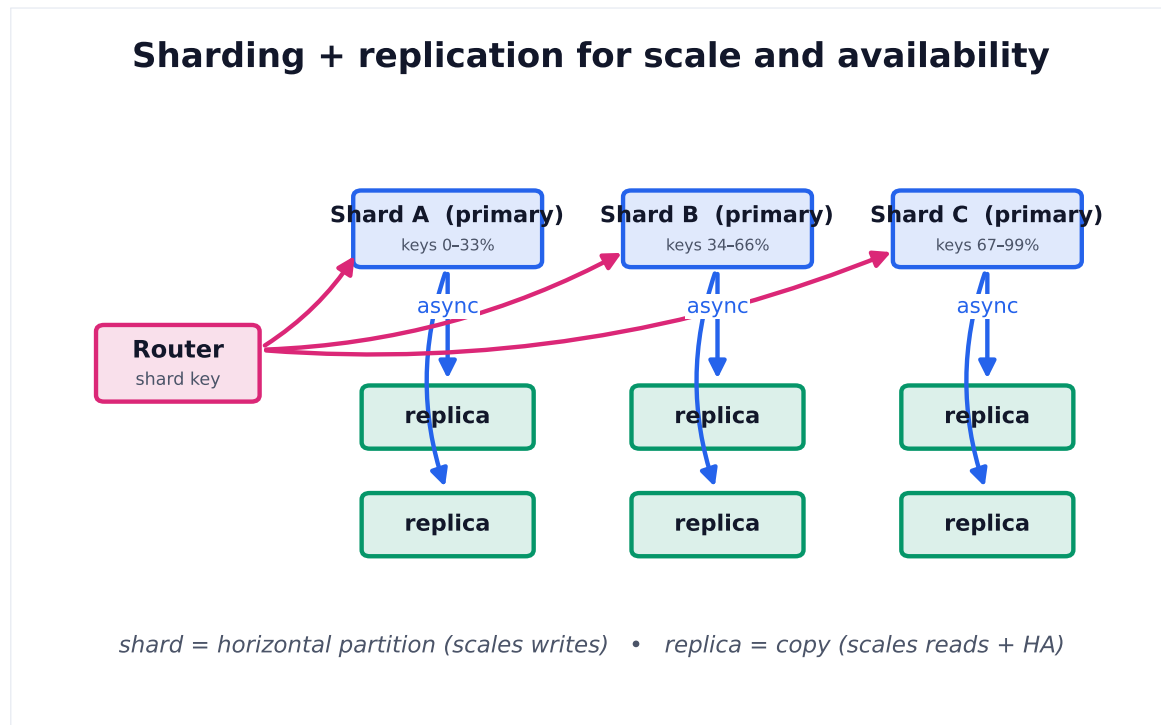


Figure 1. Sharding partitions data by key to scale writes across nodes; replication copies each shard for read scaling and failover. Most large datastores combine both.

- › Driver location also published to **Kafka** topic `driver.location` for:
 - Live map updates pushed to rider via WebSocket.
 - Surge pricing calculation.
 - ETA recalculation.
 - Trip trace storage (Cassandra consumer).

6.4 Surge Pricing

```

1 demand_cells[geohash6] = count of ride requests in last 5 min
2 supply_cells[geohash6] = count of available drivers in cell
3
4 surge_multiplier = max(1.0, demand / (supply * target_utilization_factor))

```

- › Computed every 30 s by Surge Service consuming Kafka streams.
- › Stored in Redis: `SURGE:{geohash6} → multiplier (float)`.
- › Ride estimate reads from Redis before quoting price.
- › **UI UX:** show surge visually (heatmap) and require explicit rider confirmation if multiplier > 1.5×

6.5 Trip Lifecycle --- State Machine

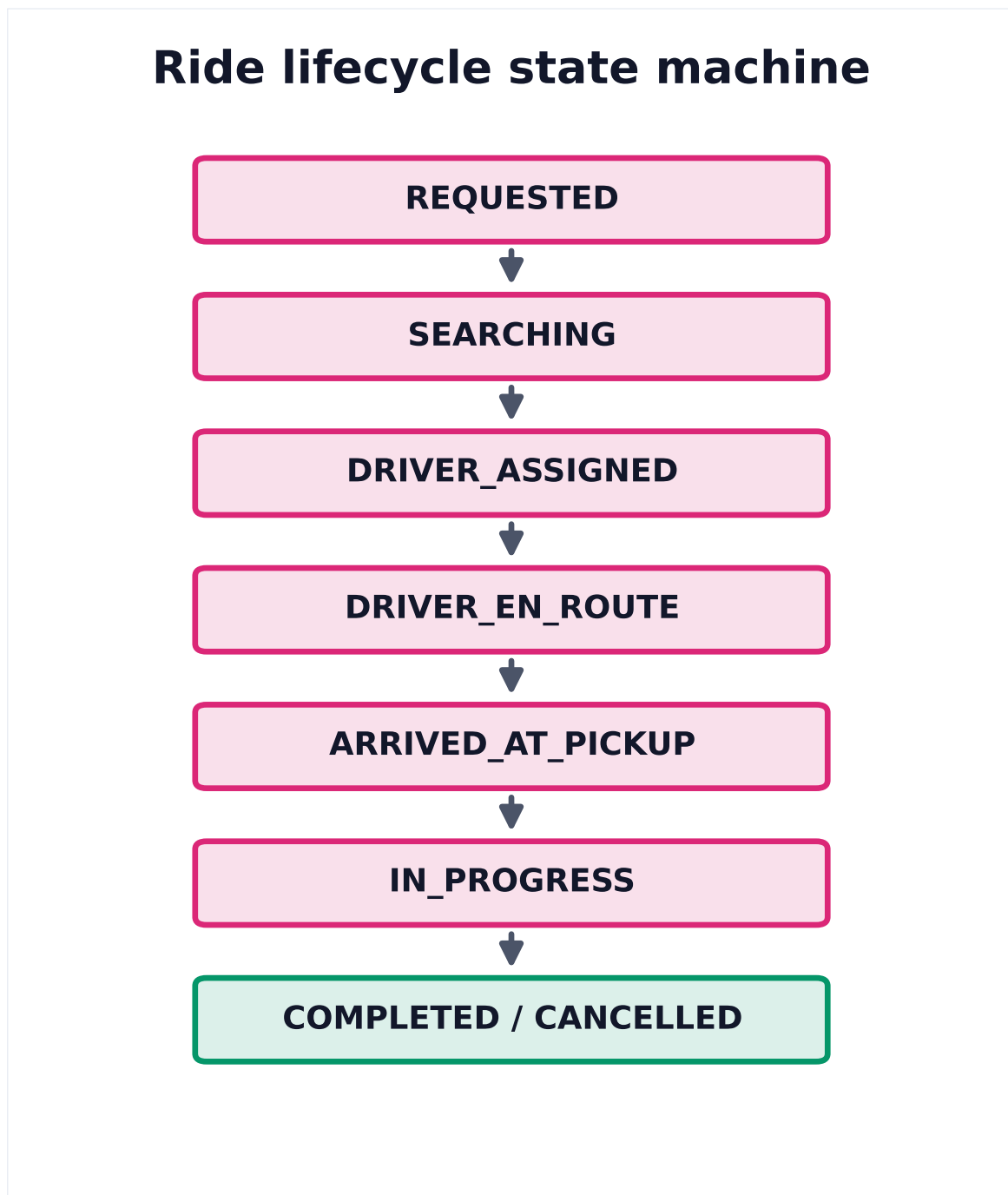


Figure 2. Modeling the ride as an explicit state machine makes valid transitions (and the events that trigger them) unambiguous — the backbone of the matching and trip services.

State transitions are persisted in Postgres with timestamps. Events published to Kafka on each transition. Kafka consumers handle: notifications, billing trigger (on COMPLETED), analytics.

6.6 ETA Calculation

- › **Routing engine:** OSRM or Google Maps Platform API call on driver assignment.
- › **Real-time recalculation:** every 30 s while driver is en route, re-query routing engine with current driver location.
- › **Pre-computed ETA surfaces:** city road graph stored in-memory (contraction hierarchies) for sub-100ms routing. Uber uses their own in-house routing (H3-based graph).

2.7 7. Bottlenecks & Scaling

Bottleneck	Mitigation
Location write storm	Redis Cluster, shard by city/geohash; accept at-most-once semantics
Matching hot cities (NYC peak)	City-level matching service instances; limit to local Redis shard
Rides DB write contention	Shard by city_id; use UUIDs to avoid hot auto-increment keys
WebSocket connection scaling	Stateful WS servers behind L4 LB; use Redis Pub/Sub as broker
Driver double-booking	Redis NX lock on driver_id during match; release on timeout
Surge pricing staleness	Short TTL (30 s) + frequent recalculation; eventual consistency OK

2.8 8. Trade-offs Summary

Decision	Option A	Option B	Choice
Geospatial index	Geohash (Redis GEODATA)	Custom Quadtree	Geohash (Redis)
Driver location transport	HTTP polling (4 s)	Persistent WebSocket	HTTP/2 (simpler, scales well)
Matching concurrency control	Postgres FOR UPDATE	Redis distributed lock	Redis lock (lower latency)
Location store	Redis in-memory	PostGIS	Redis (1M+ writes/sec)
Trip state persistence	Postgres	Cassandra	Postgres (ACID for state machines)
Routing engine	Google Maps API	Self-hosted OSRM	OSRM at scale (cost)

Interview takeaways:

- › **The geospatial indexing approach is the core of this design** — be ready to explain Geohash precision levels.
- › **Driver location is write-heavy, trip state is consistency-critical** — use different stores for each.
- › **The matching loop is an iterative fan-out** — must handle timeouts and fallbacks gracefully.
- › Surge pricing uses supply/demand ratio per geohash cell — a simple but powerful formula.

3 3. Design a Distributed Cache (like Redis/Memcached at Scale)

3.1 1. Requirements

Clarifying questions:

- › Key-value only, or also sorted sets, pub/sub, streams?
- › What are the SLAs on read/write latency?
- › Strong consistency needed, or eventual?
- › Target use case: session cache, DB cache, rate limiter, leaderboard?
- › What's the expected working set size (total hot data)?

Functional Requirements:

- › GET key, SET key value [TTL], DEL key, MGET, MSET.

- › TTL-based expiration.
- › LRU eviction when memory is full.
- › Replication for high availability.
- › Horizontal scaling via consistent hashing.
- › Pub/Sub (optional advanced feature).

Non-Functional Requirements:

- › Read latency < 1 ms p99.
- › Write latency < 2 ms p99.
- › Availability: 99.99% (cache miss falls through to DB, but should be rare).
- › No data loss for critical caches (optional persistence via WAL/RDB).
- › Scale to terabytes of cached data across a cluster.

3.2 2. Capacity Estimation

```

Cached data size (working set)   = 10 TB
Node RAM (cache node)           = 256 GB
Nodes needed                     = 10 TB / 256 GB ≈ 40 cache nodes

Read QPS                        = 1,000,000 reads/sec (1M QPS)
Write QPS                       = 100,000 writes/sec (10% of reads)

Per-node QPS (40 nodes)         = 25,000 reads + 2,500 writes per node
Single Redis node can handle    ≈ 100K-200K ops/sec → 40 nodes has 4-8× headroom

Network bandwidth per node:
  25K reads × avg 1 KB value     = 25 MB/s per node (well within 10 GbE NIC)

Memory per key overhead (Redis): ~70 bytes metadata + value size
1B keys × 70 B overhead         = 70 GB overhead in cluster (significant at scale)

```

3.3 3. API Design

Client-facing (wire protocol — Redis-compatible RESP):

```

GET <key>                → value | (nil)
SET <key> <value> [EX secs] → OK
DEL <key> [key ...]       → (integer) count deleted
MGET <key> [key ...]      → array of values
MSET <key> <val> [key val] → OK
TTL <key>                 → seconds remaining
INCR / DECR <key>         → new integer value
EXPIRE <key> <secs>       → 1 (success) | 0 (no such key)

```

Admin / Internal API:

```

CLUSTER ADDNODE <host:port> → join node to cluster
CLUSTER FAILOVER             → trigger manual leader promotion
INFO                         → stats (memory, QPS, hit_rate, evictions)
BGSAVE                       → async RDB snapshot

```

3.4 4. Data Model

In-memory data structures:

```

Hash table: key → { value, type, TTL_expiry_ts, LRU_clock }

Value types:
  String → raw bytes (up to 512 MB)
  List   → doubly-linked list (small) or ziplist (< 128 elements)

```

Hash → hash table or ziplist
 Set → hash table or intset
 ZSet → skip list + hash table

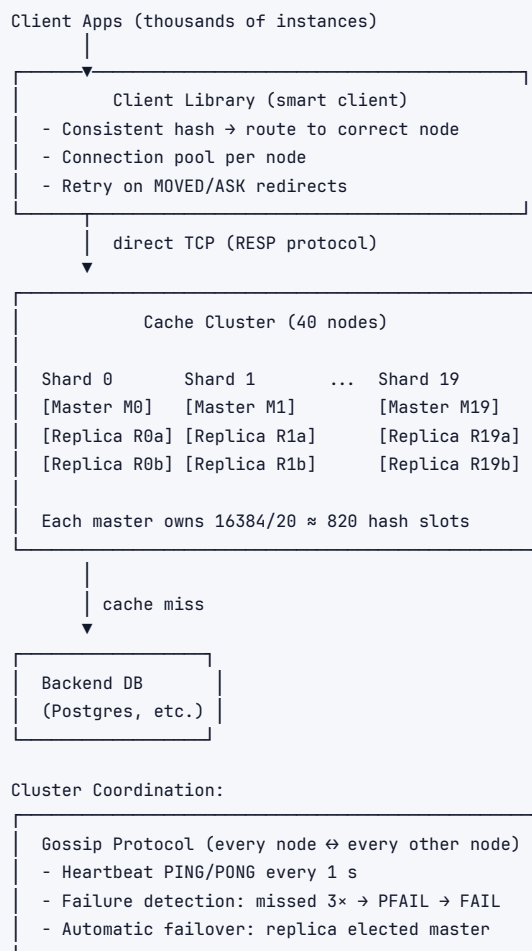
Persistence (optional):

RDB snapshot: point-in-time binary dump, every N seconds or M writes
 AOF (Append-Only File): write-ahead log, fsync policy:
 - always → 1 write per command, safest, slowest
 - everysec → fsync every 1s, at most 1s of data loss
 - never → OS decides, fastest, most data loss risk

Cluster topology metadata:

Cluster config (stored in every node):
 slot_assignments[0..16383] → node_id (Redis uses 16384 hash slots)
 node_registry: { node_id, host, port, role: master|replica, slots }

3.5 5. High-Level Architecture



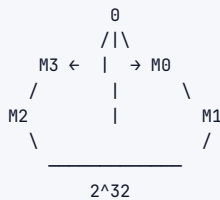
3.6 6. Deep Dives

6.1 Consistent Hashing

Problem: Naive hash(key) % N rehashes ALL keys when N changes (add/remove node).

Solution — Consistent Hash Ring:

Virtual ring: 0 to 2^{32}
 Each node placed at multiple positions (virtual nodes / vnodes)
 - 150 vnodes per physical node → even distribution
 - Rebalancing a new node: only transfer $1/N$ of keys (not all)
 key → hash(key) → position on ring → clockwise next node = owner



Key decision: 150–200 virtual nodes per physical node balances load even with heterogeneous node sizes (weight vnodes by RAM). With 40 nodes × 150 vnodes = 6,000 points on ring.

6.2 Replication

- › Each shard: 1 master + 2 replicas (RF=3).
- › **Async replication** (default Redis): master returns OK to client, replicates to replicas asynchronously. Risk: up to N ms of data loss on crash.
- › **Semi-sync (WAIT command):** WAIT 1 100 — wait for 1 replica to ack within 100ms. Better durability at slight latency cost.
- › Replicas serve **reads** to scale read-heavy workloads (with possible stale reads).
- › On master failure: replica with lowest replication lag is elected master (Raft-adjacent gossip vote).

6.3 Eviction Policies (LRU and Beyond)

Policy	Description	Use case
noeviction	Return error when memory full	Primary DB (never evict)
allkeys-lru	Evict globally least-recently-used key	General cache
volatile-lru	Evict LRU only among keys with TTL set	Mixed TTL/no-TTL data
allkeys-lfu	Evict least-frequently-used (Redis 4.0+)	Skewed access patterns
volatile-ttl	Evict key with shortest remaining TTL	Time-sensitive caching
allkeys-random	Evict random key	Uniform access, simplest

Approximate LRU (Redis implementation):

- › Full LRU requires $O(N)$ scan — too slow.
- › Redis samples 5 (configurable) random keys, evicts the least-recently-used among them.
- › At sample=10, approximates true LRU with < 1% error in most workloads.

Key decision: Use **allkeys-lfu** for a general-purpose cache — LFU outperforms LRU when there are periodic scans that pollute the LRU queue with cold data.

6.4 Cache Invalidation Strategies

Strategy	Description	Pros	Cons
TTL expiry	Set TTL on write; auto-expire	Simple	Stale window = TTL duration

Strategy	Description	Pros	Cons
Write-through	Write to cache + DB simultaneously	Always consistent	Write latency increased
Write-behind (async)	Write to cache, async flush to DB	Low write latency	Data loss risk
Cache-aside	App reads cache; miss → read DB → populate cache	DB is source of truth	Cache miss latency spike
Read-through	Cache library fetches from DB on miss automatically	Transparent to app	Library complexity
Event-driven invalidation	DB change → CDC event → DEL from cache	Real-time consistency	Infrastructure complexity

Key decision: TTL + event-driven invalidation (CDC via Debezium/Kafka). TTL covers eventual consistency; CDC invalidates on DB write for critical data.

6.5 Hot Key Problem

Problem: A single key (e.g., a celebrity's profile `user:12345`) gets 100K QPS — overwhelming the one shard that owns it.

Solutions:

1. **Key replication / local cache:** Client-side in-process cache (LRU, 100 MB per app server) for top-N hot keys. App caches key for 1 s locally.
2. **Scatter with suffix:** Store key as `user:12345:0` through `user:12345:9` (10 shards); reads pick random suffix, writes update all. Works for read-heavy hot keys.
3. **Read replicas:** Route reads to any of the replicas for that shard.
4. **Detect & alert:** Monitor per-key QPS in the client library; auto-promote to local cache.

6.6 Cache Stampede (Thundering Herd)

Problem: Popular key expires → 10K concurrent requests all miss → all hit DB simultaneously → DB overloaded.

Solutions:

1. **Probabilistic early expiration (PER):** Recompute cache key slightly before TTL expires (during window $TTL - \text{random}(0, \beta \times \log(\text{compute_time}))$). Only one client recomputes at a time; others still serve stale value.
2. **Mutex / lock:** First cache-miss acquires a distributed lock (Redis `SET lock:key NX EX 5`). Others wait or serve stale.
3. **Background refresh:** Separate async job refreshes popular keys before they expire.
4. **Stale-while-revalidate:** Serve stale immediately; trigger async refresh; update cache when done.

Key decision: Stale-while-revalidate for most use cases — zero extra latency for users; background refresh handles expiry.

6.7 Write Policies

Write-Through:	Client → Cache & DB (synchronous both) + Consistent - Higher write latency
Write-Back:	Client → Cache (immediate) → DB (async batch) + Low latency - Risk of data loss
Cache-Aside:	Client → DB directly; read-path populates cache + DB authoritative - Cold start cache misses

6.8 Node Failure Handling

- › **Detection:** Gossip heartbeat; node marked PFAIL after missing 3 pings; cluster votes to mark FAIL (majority of masters agree).
- › **Failover:** Replica with smallest replication offset (least lag) wins election; becomes new master for the slot range within ~1–3 s.
- › **Client impact:** During failover window, affected slots return CLUSTERDOWN errors → client retries with backoff.
- › **Split brain prevention:** Requires majority of masters alive to accept writes. With 20 masters, lose 10 → writes rejected (availability vs consistency trade-off).

3.7 7. Bottlenecks & Scaling

Bottleneck	Mitigation
Hot key (single shard 100K QPS)	Key fan-out / local in-process cache
Memory exhaustion	Eviction policy; monitor <code>used_memory_rss</code> ; add nodes
Network bandwidth (large values)	Compress values at client (snappy/lz4); max value size policy
Failover latency (1–3 s)	Keepalive + health checks; pre-elect backup masters
Replication lag	<code>WAIT 1 100</code> for critical writes; monitor <code>repl_lag</code> metric
Cross-datacenter replication	Redis Cluster doesn't span DCs natively; use CRDT-based solutions or geo-replicas with eventual consistency

3.8 8. Trade-offs Summary

Decision	Option A	Option B	Choice
Sharding strategy	Consistent hashing	Range-based sharding	Consistent hashing (smooth rebalance)
Eviction	LRU	LFU	LFU (scan-resistant)
Cache invalidation	TTL only	TTL + CDC events	TTL + CDC (real-time for critical)
Write policy	Write-through	Cache-aside	Cache-aside (DB authoritative)
Hot key	Replica reads	Client-side local cache	Local cache (zero network)
Stampede protection	Mutex lock	Stale-while-revalidate	SWR (no latency penalty)
Persistence	RDB only	AOF everysec	AOF everysec (< 1s loss)

Interview takeaways:

- › **Consistent hashing with virtual nodes is essential** — explain the 150-vnode trick for even distribution.
- › **Hot key and cache stampede are the two most common follow-up questions** — have concrete solutions ready.
- › **Redis uses approximate LRU (sampling)** — not true LRU. Know why.
- › Cache-aside is the most common pattern; write-through for critical financial data.

4. Design a Web Crawler (Google-Scale)

4.1 1. Requirements

Clarifying questions:

- › Full web crawl (all URLs), or focused crawl (specific domains/topics)?
- › How fresh does the data need to be? (daily recrawl vs. weekly vs. on-change)
- › Should we store raw HTML or parsed content?
- › Do we need to handle JavaScript-rendered pages (dynamic content)?
- › What's the target crawl rate and storage budget?

Functional Requirements:

- › Start from a set of seed URLs; discover new URLs via link extraction.
- › Download and store page content.
- › Respect robots.txt and crawl-delay directives.
- › Deduplicate URLs — don't crawl the same page twice (in a given window).
- › Freshness — recrawl pages periodically based on update frequency.
- › Output: raw HTML (and/or extracted text) + metadata per URL.

Non-Functional Requirements:

- › Scale: crawl 10 billion pages (Common Crawl scale).
- › Throughput: 1,000 pages/sec sustained (86M pages/day → 10B pages in ~116 days).
- › Distributed: hundreds of crawler nodes.
- › Politeness: max 1 req/sec per domain; honor robots.txt.
- › Durability: don't re-crawl already-crawled URLs unless recrawl is due.
- › Freshness: high-value pages (news) re-crawled hourly; static pages monthly.

4.2 2. Capacity Estimation

```

Target pages           = 10 billion
Crawl rate             = 1,000 pages/sec
Time to crawl all      = 10B / 1,000 = 10,000,000 s ≈ 116 days (1 full crawl cycle)
Peak rate needed       = 3,000-5,000 pages/sec (to finish faster / handle crawl failures)

Avg page size          = 100 KB HTML
Storage (HTML only)    = 10B × 100 KB = 1 PB
Compressed (~5:1 gzip) = ~200 TB

Links extracted per page = avg 50 outgoing links
Total URL queue entries = 10B × 50 = 500B (dedup reduces this dramatically)
Seen URL set (dedup)    = 10B URLs × ~100 bytes/URL = 1 TB (in memory: impractical)
                        Bloom filter: 10B items, 1% FP → ~12 GB (fits in RAM!)

DNS lookups:
1,000 pages/sec; avg 50 unique domains/sec
DNS cache (TTL=300s) covers most; ~5 uncached DNS lookups/sec

```

4.3 3. API Design

Internal service APIs (not user-facing):

```

1 # URL Frontier Service
2 POST    /frontier/enqueue
3         Body: { url, priority, source_page, discovered_at }

```

```

4     Returns: { queued: true | false (already seen) }
5
6 GET   /frontier/next?crawler_id=&batch_size=100
7     Returns: [{ url, fetch_after_ts }] (respects politeness delay)
8
9 POST  /frontier/done
10     Body: { url, status_code, crawled_at, content_hash }
11
12 # Content Store
13 PUT   /content/{url_hash}
14     Body: { raw_html, headers, crawled_at, content_hash }
15
16 GET   /content/{url_hash}
17
18 # Robots.txt Cache
19 GET   /robots?domain=example.com
20     Returns: { allowed_paths, disallowed_paths, crawl_delay_s, sitemap_urls }
21
22 # Recrawl Scheduler
23 POST  /recrawl/schedule
24     Body: { url, next_crawl_at, priority }

```

4.4 4. Data Model

URL Frontier (Priority Queue + Politeness Queue):

```

1 frontier_urls (
2   url_hash      BIGINT PRIMARY KEY,  -- xxHash64 of normalized URL
3   url           TEXT,
4   priority      FLOAT,               -- higher = crawl sooner
5   domain        TEXT,
6   next_fetch_ts TIMESTAMPTZ,        -- earliest time to fetch (politeness)
7   status        ENUM('pending', 'in_flight', 'done', 'error'),
8   retry_count   INT DEFAULT 0
9 )
10 Implemented as: Apache Kafka topics per priority band (high/medium/low)
11                + Redis ZSETs per domain for politeness scheduling

```

Crawled Pages (blob + metadata):

```

pages (
  url_hash      BIGINT PRIMARY KEY,
  url           TEXT,
  domain        TEXT,
  content_hash   BIGINT,             -- SimHash for near-duplicate detection
  crawled_at    TIMESTAMPTZ,
  http_status    INT,
  content_type   TEXT,
  raw_html_key   TEXT               -- S3 key for raw HTML blob
)

```

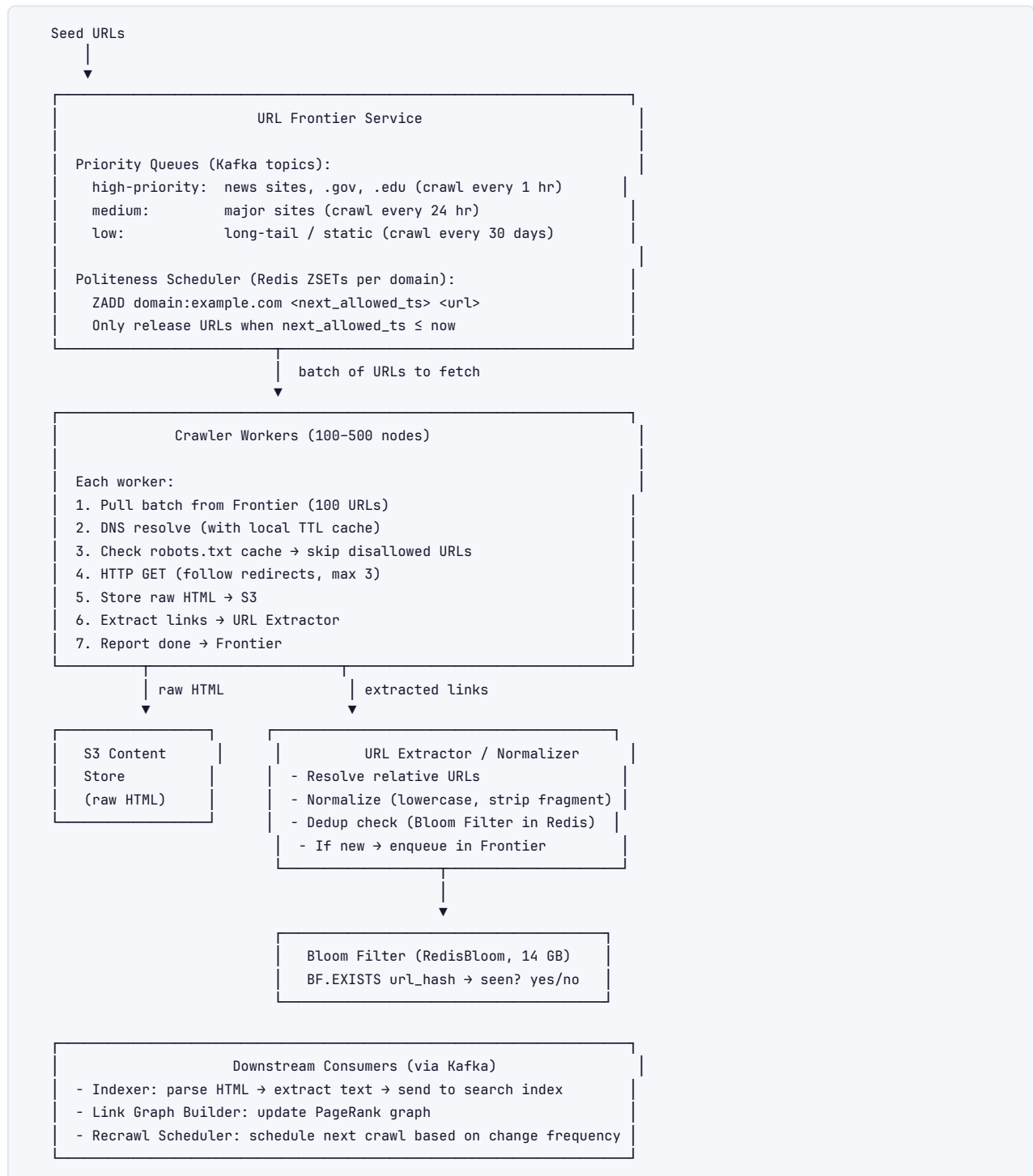
Seen URL Set (deduplication):

Bloom Filter (in-memory, distributed):

- 10B URLs, 0.1% false positive rate → ~14.4 GB
- Implementation: Redis BF.ADD / BF.EXISTS (RedisBloom module)
- Fallback: persistent seen_urls table in Cassandra for exact check on BF hit

robots.txt Cache:

robots_cache (domain → RobotsTxt object, TTL=24h) in Redis

4.5 5. High-Level Architecture**4.6 6. Deep Dives****6.1 URL Frontier --- Priority + Politeness****Two-tier queue architecture (Mercator design):**

```

Front-end queues (priority):      Back-end queues (politeness):
Q_high [news, trending]          Domain A: [url1, url3, url7] → min 1s delay
Q_med  [general web]             Domain B: [url2, url5]   → min 1s delay
Q_low  [static, rarely updated]  Domain C: [url4, url6]   → min 5s delay

Selector: reads from front-end queues proportionally to priority.
          Routes each URL to its domain back-end queue.
          Back-end queue dispatcher waits for per-domain cooldown before releasing.

```

Priority assignment:

- › PageRank estimate of source page.
- › Domain authority (Alexa/Majestic rank).
- › Time since last crawl vs. estimated update frequency.
- › Explicit signals (sitemap priority, news feeds).

Implementation:

- › Priority bands implemented as **Kafka topics** (`crawl.high`, `crawl.med`, `crawl.low`).
- › Politeness tracking: **Redis sorted set** per domain — `ZADD domain:example.com {next_fetch_ts} {url}`. Poller does `ZRANGEBYSCORE domain:* -inf now LIMIT 10` every second.

6.2 URL Deduplication --- Bloom Filter vs Exact Set

Approach	Storage for 10B URLs	Lookup time	False positives	False negatives
Bloom Filter (1% FPR)	~12 GB	O(k) hashes	1% (skip new URL)	0%
Bloom Filter (0.1% FPR)	~14.4 GB	O(k) hashes	0.1%	0%
Hash Set (in-memory)	~1 TB (impractical)	O(1)	0%	0%
Cassandra exact store	~200 GB on disk	1–5 ms	0%	0%
Redis Set	~600 GB	O(1)	0%	0%

Key decision: Bloom Filter as first-pass dedup (RedisBloom, ~14 GB), with Cassandra exact-set for conflict resolution on BF hit (false positive check).

- › BF says "seen" → check Cassandra exact table (1% of traffic → Cassandra QPS manageable).
- › BF says "not seen" → guaranteed new URL → enqueue directly.

URL normalization before hashing:

```

1. Lowercase scheme and host
2. Remove default port (:80, :443)
3. Sort query parameters alphabetically
4. Remove URL fragment (#anchor)
5. Resolve relative paths (/../ → /)
6. Strip tracking params (utm_source, fbclid, etc.)
7. Canonicalize: prefer canonical link rel tag in HTML

```

6.3 DNS Resolution

Problem: 1,000 pages/sec from 1,000 different domains = 1,000 DNS lookups/sec. Public DNS resolvers (8.8.8.8) rate-limit at ~few hundred QPS.

Solutions:

1. **Local DNS cache per crawler node:** TTL-respecting LRU cache. Cache hit rate > 90% for popular domains.
2. **Dedicated DNS resolver cluster:** Internal recursive DNS servers (unbound / BIND), scaled horizontally. Query rate: ~50 uncached lookups/sec per node.
3. **Prefetch DNS:** When a new domain is discovered, pre-resolve asynchronously before it's scheduled for crawl.
4. **DNS blacklist:** Skip private IPs (SSRF protection: 10.0.0.0/8, 192.168.0.0/16, 127.0.0.1).

6.4 robots.txt Compliance

```
For domain example.com:
1. Check Redis cache: GET robots:example.com
2. Cache miss → fetch https://example.com/robots.txt
3. Parse:
    User-agent: *
    Disallow: /private/
    Crawl-delay: 5
    Sitemap: https://example.com/sitemap.xml
4. Store in Redis: SET robots:example.com <parsed_rules> EX 86400
5. Before each URL fetch: check if path matches Disallow patterns

Politeness: max(1s_default, Crawl-delay_from_robots)
```

Gotchas:

- › robots.txt itself may not exist (HTTP 404 → treat as "allow all").
- › robots.txt may be too large (truncate at 512 KB).
- › Different user-agent rules: use "Googlebot" or your crawler's name.

6.5 Trap Avoidance

Spider traps generate infinite URLs (e.g., /calendar/next?date=... → infinite future dates).

Mitigation:

1. **URL depth limit:** max 10 hops from seed URL.
2. **Per-domain URL count limit:** max 100K URLs per domain (override for news sites).
3. **URL path pattern detection:** if same path template seen > 1,000 times, blacklist pattern.
4. **Cycle detection:** if URL hash appears in path history (seen in URL chain) → stop.
5. **Content hash dedup:** if page content SimHash is near-duplicate of already-crawled page → skip storing, but do extract links (might have new ones).

6.6 Near-Duplicate Detection (SimHash)

```
1. Extract tokens (words) from page HTML.
2. For each token, compute hash (64-bit).
3. For each bit position 0..63:
    if token_hash bit is 1 → add weight to column[bit]
    if token_hash bit is 0 → subtract weight
4. Final SimHash: bit[i] = 1 if column[i] > 0, else 0.

Two pages with SimHash Hamming distance ≤ 3 → near-duplicate.
Storage: 8 bytes per page × 10B pages = 80 GB → manageable in distributed store.
```

6.7 Distributed Coordination

Problem: 500 crawler nodes must not fetch the same URL concurrently.

Solution:

- › URL Frontier is the single coordinator.
- › Each URL assigned to one crawler node (round-robin or URL-hash based routing).

- › Frontier marks URL `in_flight` on dispatch; reverts to pending if heartbeat missing for 60 s (dead crawler).
- › No two nodes ever assigned the same URL from the frontier (frontier is authoritative).

Crawler node health:

- › Periodic heartbeat to Frontier Service (every 10 s).
- › Dead node's `in_flight` URLs requeued automatically after 60 s timeout.

6.8 Freshness & Recrawl Scheduling

Page type	Crawl frequency	Signal
News article (first 1h)	Every 15 min	Source is major news domain + recent publish
News article (1–24h)	Every 2 hr	Declining share velocity
Popular blog post	Daily	Sitemap <code>changefreq=daily</code>
Product page	Weekly	Sitemap <code>changefreq=weekly</code>
Static reference page	Monthly	Sitemap <code>changefreq=monthly</code> or inferred
Newly discovered URL	Once (ASAP)	Not yet crawled

Adaptive recrawl:

- › Track $\text{change_rate} = \text{content_hash_changes} / \text{recrawls per URL}$.
- › Increase frequency if page changes often; decrease if rarely changes.
- › $\text{next_crawl_delay} = \max(\text{min_delay}, \text{last_crawl_interval} \times (1 - \text{change_rate} + 0.1))$

4.7 7. Bottlenecks & Scaling

Bottleneck	Mitigation
URL Frontier throughput	Partition Kafka topics by domain-hash; horizontal Frontier shards
Bloom filter accuracy drift	Use Counting Bloom Filter or periodic rebuild; monitor FP rate
DNS resolution rate	Dedicated internal DNS cluster + per-node LRU cache
S3 write throughput	Randomize S3 key prefix; use multipart upload for large HTML batches
robots.txt fetch overhead	Redis cache (TTL=24h); prefetch on domain discovery
Spider traps infinite loop	URL depth limit + per-domain URL count cap
Crawler node failure	Frontier requeues <code>in_flight</code> URLs after heartbeat timeout
Hot domains (Reddit, Twitter)	Per-domain rate limit enforced strictly; queue backs up, not domain

4.8 8. Trade-offs Summary

Decision	Option A	Option B	Choice
URL dedup	Exact hash set (Cassandra)	Bloom filter + exact fallback	BF + exact fallback (14 GB vs 200 GB)
URL frontier storage	Database (Postgres)	Kafka topics + Redis ZSETs	Kafka + Redis (throughput)

Decision	Option A	Option B	Choice
DNS	Public resolvers (8.8.8.8)	Internal recursive DNS cluster	Internal DNS (rate limit avoidance)
Content storage	HDFS	S3	S3 (elastic, cheap, managed)
Near-duplicate detection	Full content MD5 comparison	SimHash (64-bit)	SimHash (O(1) compare, 8B/page)
JS rendering	Skip JS pages	Headless Chrome (Puppeteer)	Skip first pass (cost); headless for top-N sites
Crawl scheduling	Fixed interval	Adaptive (change-rate based)	Adaptive (better freshness/cost ratio)

Interview takeaways:

- › **The URL Frontier is the heart of the crawler** — the two-tier priority + politeness design (Mercator) is the canonical answer.
- › **Bloom filter for dedup at 10B scale** — know the math: 10B items, 0.1% FPR → 14.4 GB. This is the key insight.
- › **Politeness is non-negotiable** — not just ethical; aggressive crawling gets IP-banned, defeating the crawler.
- › **Distributed coordination:** URL Frontier as single source of truth prevents duplicate crawling across nodes.
- › **Spider trap avoidance:** URL depth limit + per-domain URL count cap are the two simple, effective defenses.

End of Part 2 — Advanced System Design Problems